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# Precipitate evolution in the early stages of ageing in Inconel 718 investigated using small-angle x-ray scattering

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ABSTRACT

Microstructural evolution during the early stages of ageing (less than one hour) in a Ni–Cr–Fe based superalloy Inconel 718 (IN718) has been investigated using Small-Angle X-ray Scattering (SAXS). The effects of precipitate kinetics on the precipitate size distribution are compared indirectly with SAXS measurements by using Vickers microhardness data. The microhardness increased after 4 min of ageing at a temperature of 760 °C, although the recorded SAXS data did not reveal the precipitate size distribution. This indicates that the precipitates had not evolved enough to be detected, but still a small number of precipitates increased the yield strength. After ageing the alloy for the shortest period for which data were available, 8 min, clear evidence of precipitates could be found from the SAXS data, showing that the $\gamma''-$ precipitates are about 6 nm in width and 3 nm in height.

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## 1. Introduction

Excellent mechanical strength and resistance to creep at elevated temperatures are important requirements in today's aircraft engines. Over the years, special kinds of alloys have been developed, often referred to as superalloys, which fulfil these requirements. The Ni–Cr–Fe based superalloy Inconel 718 (IN718) possesses excellent mechanical strength and resistance to creep at operating temperatures up to approximately 650 °C. These attributes make the alloy suitable for use in aero engines and power generation turbines [1]. IN718 is a precipitate hardening alloy. The two precipitates that contribute to increased hardness are the ordered face centre cubic (fcc) $\gamma'-$phase of composition $Ni_3$(Al,Ti), and the coherent ordered disc-shaped body-centred tetragonal (bct) $\gamma''-$phase, which comprises nickel, niobium, titanium and aluminium $Ni_3$(Nb, Ti, Al) [2]. When the IN718 alloy has been aged past its nucleation stage, approximately 13 vol% of $\gamma''$ and 4 vol% of $\gamma'$ precipitates are present in the matrix [3]. The size of the precipitates depends on the ageing temperature and time of heat treatment, and has been measured by a number of researchers for ageing times longer than two hours, e.g. Camus and Engberg [4], Han et al. [5,6], Sundararaman et al. [7] and Slama et al. [8,9].

Transmission Electron Microscopy (TEM) dark field techniques have been the dominate method of studying nucleation and growth of $\gamma'$ and $\gamma''$ precipitates. In the study by Alam et al. [10], WE91 (a variant of IN718) was investigated for very short ageing times. They utilised both TEM and atom probe tomography and compared Vickers microhardness data to the microstructural information. After ageing for 120 s, they found a small increase in microhardness, but no evidence of nucleation in the form of superlattice reflections. After 500 s evidence of superlattice reflection could be found. Prior to the observation of precipitates, clustering of aluminium/titanium and niobium has been shown to occur, which is believed to contribute to an increase in hardness. The ageing temperature was 706 °C. Nakai et al. [11] investigated the effects of solution treatment on microstructural evolution in the early stage of ageing. They used the Vickers microhardness technique to measure the hardness variation with ageing time, where the shortest ageing time was 10 s.

The morphology and distribution of stable precipitates in IN718 is well understood. However, less is known about the nucleation and growth stage of these precipitates. Measurements of the

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nucleation and growth processes in engineering alloys are valuable for calibrating and validating material models [12]. In this study, we consider measurements of alloy IN718 in the early stages of nucleation and growth using Small-Angle X-ray Scattering (SAXS) and Vickers microhardness data for comparison.

## 2. Experimental methods

Aged heat treated samples were investigated using the Vickers microhardness test method [13]. The samples were first hot rolled and solution treated at 954 °C for 1 h. Following this, ageing was carried out at eight temperatures (500, 550, 600, 650, 700, 760, 800 and 850 °C), each for eight different times (4, 8, 16, 32, 64, 120, 240 and 480 min), giving 64 samples. Each of the 64 aged samples was etched lightly in oxalic acid with a voltage of ~3 V to enable Vickers microhardness testing on a single grain. In order to obtain significant statistics, five measurements for each specimen were performed using a load of 200 g and a 15 s dwell time.

SAXS is a non-destructive technique for obtaining quantitative information such as shape and size distribution of clustering of elements that have high electron density contrast with respect to the matrix [14-17]. In IN718, $\gamma'$ particles comprise nickel, aluminium and titanium, and $\gamma''$ nickel, niobium, titanium and aluminium, respectively. The chemical composition of alloy IN718 is given in Table 1. It is possible impurities segregate in the alloy, causing false results from the scattering of the x-ray beam. Since the alloying elements in superalloys are very controlled, only a small amount of impurities are expected. Because of this, the authors do not expect any large amount of impurities, which can segregate and change the SAXS-curves in the $q$-range used in this work.

For a fully aged IN718 alloy, the volume fraction ratio of $\gamma''/\gamma'$ precipitates is about three [3]. Alam et al. [10] reported that the volume fraction ratio of the precipitates is higher for shorter ageing times (in their case 500 s for ageing at 706 °C) using Atom Probe Tomography (APT). Also, they reported that the $\gamma'$ and $\gamma''$ precipitates are located in the same regions, and very often as co-precipitates forming a sandwich-like structure of the form $\gamma'/\gamma''/\gamma'$ or $\gamma''/\gamma'/\gamma''$.

From this knowledge, it is believed that SAXS can be used to predict the evolution of $\gamma''$ precipitates during ageing.

SAXS measurements were carried out at the synchrotron research facility MAX IV Laboratory located in Lund, Sweden, on the I911-4 SAXS beamline [18], at a fixed wavelength of 0.91 Å. The accessible scattering vector $q$-range was [0.01, 0.3] Å$^{-1}$, and the beam size was 200 × 300 μm. The $q$-range allows us to resolve particles between ~10 and 300 Å. SAXS patterns were acquired using a Pilatus 1 M two-dimensional detector with an exposure time of 150 s for each frame, normalised by the incoming photon flux and sample transmission. The hot rolled samples, with a thickness of ~40 μm, were first solution heat treated, and then subsequently aged at 760 °C for 4, 8, 16, 32 and 64 min. Details of how the precipitate size was extracted are given in Section 3.

**Table 1**
Chemical composition of alloy IN718 (wt%).

| Ni | Cr | Fe | Nb | Mo | Ti | Al | Co |
|---|---|---|---|---|---|---|---|
| 53.53 | 18.41 | 17.92 | 5.00 | 3.05 | 0.94 | 0.6 | 0.17 |
| C | Mn | Si | P | S | B | Cu | |
| 0.03 | 0.11 | 0.08 | 0.010 | 0.0004 | 0.002 | 0.13 | |

## 3. SAXS raw data

The inset graph in Fig. 1(a) shows the 2D SAXS pattern of a sample aged for 32 min, while the main graph shows the magnification of the pattern close to the beam stop. The colour scale represents the SAXS intensity for each pixel. The black horizontal and vertical lines are caused by the detector gaps. The larger black area in the middle is the beam stop. The structure of the 2D SAXS pattern shows a hexagonal shape, which indicate streaks in the projection in various directions. In the literature four different origins of streaks in SAXS patterns are reported: (I) presence of an oxide layer on the surface of the specimen [19,20], (II) presence of aligned anisotropically shaped precipitates [21,22], (III) possible double Bragg reflections within a single grain [23,24], or (IV) presence of ordered dislocations [25-27]. In further analysis, the main streaks are believed to arise from either (II) or (IV) as IN718 is generally resistant to oxidation, and the double Bragg reflections are assumed to be neglected.

To analyse the anisotropic behaviour, the intensity variation as a function of azimuthal angle $\varphi$ was obtained for radius $r_\varphi$, which is illustrated in Fig. 1(b). The azimuthal intensity I($\varphi$, $t$=32 min) shows symmetric behaviour, containing peaks and valleys (solid line). The large peak at $\varphi \approx 60°$ disappears (dashed line) after subtracting the azimuthal intensity of the 4 min aged sample (without precipitates). Therefore, it can be assumed the streaks observed in the SAXS pattern (solid line) arise from structural anisotropy in the matrix, e.g. ordered dislocations, and not from the presence of aligned anisotropically shaped precipitates. The variation observed in the subtracted azimuthal intensity plot (dashed line) is therefore believed to arise from sample inhomogeneities. If aligned precipitates are present, distinct and symmetrical peaks should be observed in the subtracted spectra. The present peaks can be assumed to arise from the rolling direction during manufacturing and changes during heating.

From the analysis of the streaks above, the conclusion is that the precipitates are randomly oriented in the illuminated volume. With this knowledge, intensity plots are constructed by radial averaging of the data from the 2D SAXS plot and subtracting the constant value related to background scattering [28]. The curves contain information about the particle shape and size distribution.

Based on the particle shape (e.g. oblate spheroid) and the assumption of a given size distribution (e.g. log-normal), the scattering pattern can be fitted with a calculated intensity. Applying a size distribution $p(r)$ and neglecting the particle-to-particle scattering structure factor, the intensity $I$ can be described as [29]:

$$I(q) = N|\Delta\rho|^2 \int_0^\infty |V_p(r)F_p(q,r)|^2 p(r)\, dr, \tag{1}$$

where $F_p(q,r)$ is the dimensionless form factor of a particle with radius $r$ and volume $V_p(r)$ with a scattering vector $|\overline{q}| = q$. $|\Delta\rho|$ is the difference in scattering length density between the matrix and the precipitate, and $N$ is the density of precipitates (number/m$^3$). Because the SAXS intensities are not absolute differential scattering cross sections, the density parameter $N$ is a scaling factor only. The form factor of an oblate spheroid is given in [30].

For small values of $q$, Eq. (1) can be simplified using the Guinier approximation [31]:

$$I(q) \propto \exp\left(-\frac{q^2 R_g^2}{3}\right), \tag{2}$$

where $R_g$ is the radius of gyration of the overall particle distribution. This is determined from the slope of the plot of ln($I$) versus $q^2$ as $q$ tends to zero (the Guinier plot). The relation between the radius of gyration and the size of the particle for an oblate

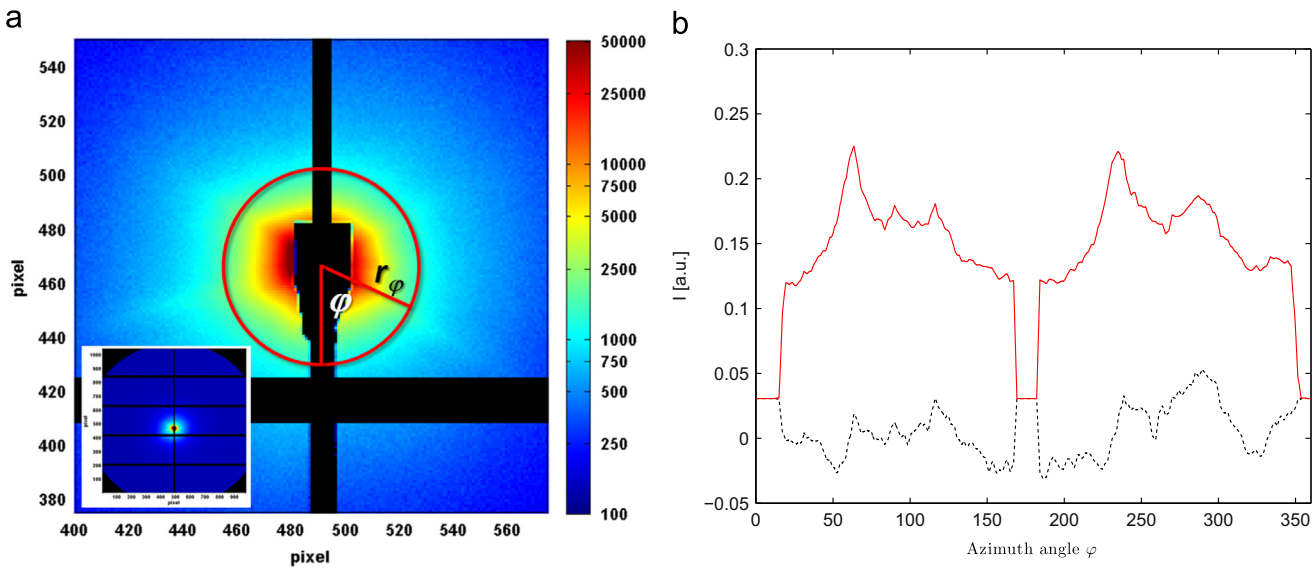


**Fig. 1.** Measured intensity distribution for a sample aged for 32 min. (a) shows the two-dimensional SAXS intensity. The colour scale represents the SAXS intensity for each pixel. (b) shows the intensity as function of the azimuthal angle $\varphi$ for a radius $r_\varphi$ (marked in Fig a). The solid line shows the raw data from (a) and the dashed line represents the same data with 4 min of ageing subtracted. (For interpretation of the references to colour in this figure caption, the reader is referred to the web version of this paper.)

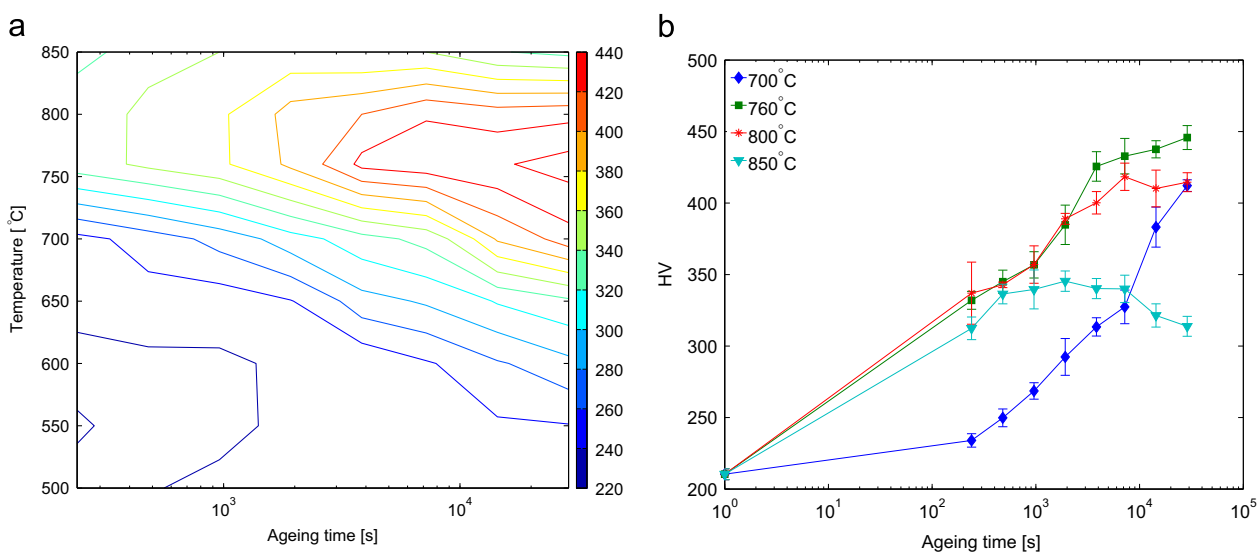


**Fig. 2.** Results of the Vickers microhardness measurements. (a) shows a contour plot of ageing time, temperature and microhardness. (b) shows the variation in microhardness with ageing time at four different temperatures. The error bars are one standard deviation of five hardness measurements.

spheroid can be expressed as [31]

$$R_g^2 = \frac{2L^2 + h^2}{20} \tag{3}$$

where $L$ and $h$ are the width and height of the precipitate, respectively.

It is known that the aspect ratio $\varepsilon = L/h$ of the $\gamma''$ precipitates in IN718 increases with increasing size [32]. Basically, the $\gamma''$ particles can be divided into two different groups according to their sizes: in the smaller group ($L < 32$ nm) the aspect ratio grows more rapidly than in the larger group. The change in growth rate is correlated with the loss of coherency as the particles become larger. In the work of Devaux et al. [32], the inverse of the aspect ratio is plotted as a function of precipitate diameter using 35 data points, clearly showing two regions with different rates of growth. Based on this plot, two equations can be constructed in order to determine the aspect ratio, depending on precipitate size:

$$\varepsilon = \begin{cases} \left(0.55 - \frac{0.185}{20}L\right)^{-1} & \text{when } 6 < L \le 32 \text{ nm} \\ \left(0.28 - \frac{0.05}{60}L\right)^{-1} & \text{when } L > 32 \text{ nm.} \end{cases} \tag{4}$$

All obtained results were computed using the SASfit software [33]. This includes both the fitted curves, and the slope of the Guinier plot. The determined value of $R_g$ is used to iteratively calculate the precipitate width and height using Eqs. (3) and (4), respectively.

## 4. Results and discussion

The following section describes the results from the SAXS and Vickers microhardness measurements. Finally, the data obtained

are compared with previously published results and a numerical particle nucleation, growth and coarsening precipitate model.

### 4.1. Vickers microhardness

The hardness of precipitate hardening alloys, such as IN718, is related to the volume fraction, composition and size of strengthening precipitates. The size and composition of the precipitates depends on the ageing temperature and time [34]. Thus, the hardness of IN718 is expected to vary for different temperatures and ageing times. The results from the microhardness study are shown in Fig. 2. Fig. 2(a) illustrates the measured hardness as a function of temperature and ageing time, and Fig. 2(b) shows the evolution of Vickers microhardness with ageing time at four temperatures: 700, 760, 800 and 850 °C. Samples aged at 760 °C show an increase in hardness for each measured time, and have, together with those aged at 700 °C, not achieved their maximum hardness after 8 h. The peak nucleation rate can be assumed to occur at this temperature and motivates the use of samples heat treated at 760 °C for the SAXS measurements.

The chemical composition and nucleation of secondary phases at the atomic level can be evaluated using APT. APT has successfully been used to measure the evolution of precipitates in the early stages of ageing for IN718 [9]. However, these measurements did not measure the actual composition of the precipitates, which affect the hardness. Similar studies, using APT, have been conducted for aluminium alloy Al–Sc–Zr [35,36]. The results showed that in the first stage a Sc enriched nucleus was formed, which was followed by a outer shell formation of Zr. This outer shell provides a secondary strength increase to the alloy [35,36]. To the authors knowledge, no such studies have been performed for IN718, but the same phenomena may exist.

### 4.2. Precipitate characterisation

Based on the results of the Vickers microhardness study, the samples heat treated at 760 °C were chosen for the SAXS measurements. Data recorded for five ageing times were analysed using SASfit following the methods described in Section 3. Fig. 3 (a) shows the resulting radially averaged intensity as a function of the scattering vector for each of the ageing times. For the shortest ageing time (4 min) the intensity distribution appears to follow a linear dependency in a log–log plot, whereas for measurement with increasing ageing times an enhancement of the intensity is observed as a "bump" in the curves. The size and $q$-value of this perturbation increases with ageing time, which is expected because of the formation of larger precipitates in the alloy. However, it is noted that the trend is not revealed in data obtained for the 16 min ageing time sample. The reason for this is believed to arise from faulty and/or missed registered data entries during the experiments.

Each of the extracted intensity distributions was fitted using the SASfit software. The solid line in Fig. 3(b) illustrates such a fit for one of the samples (32 min ageing time). Similar fits were performed for the other measurements and extracted parameters were recorded and used for further analysis. Values for $R_g$ are listed in Table 2.

A second approach to extract information about $R_g$ uses the Guinier approximation as discussed in Section 3. Fig. 4 shows the Guinier plots for the five ageing times measured at 760 °C. The observed difference in shape between the curves follows a trend strongly correlated with changes in $R_g$. Utilising linear fits to the distributions in the region $q^2 \sim 0.005$ Å$^{-2}$ allows the $R_g$ data to be extracted. Resulting values are listed in Table 2.

In order to obtain the height $h$ and the width $L$ of the precipitates from $R_g$, Eq. (3) is used in conjunction with Eq. (4). It should be noted, since the aspect ratio $\varepsilon$ depends on the precipitate width, Eq. (3) must be solved iteratively using Eq. (4). This method was utilised for the four data sets measured at 8, 16, 32 and 64 min. Resulting values are given in Table 2.

To illustrate the trend in the volume fraction evolution of the precipitates, Kratky plots ($Iq^2$ vs. $q$) [14] have been generated using SASfit software. Fig. 5 shows the Kratky representation of the SAXS data for the five ageing times measured at 760 °C. A clear bump in the distributions for longer ageing times can be observed, where

**Table 2**
Extracted $R_g$ values for the various ageing times together with corresponding calculated $\gamma''$ height ($h$) and width ($L$) (Further details in text).

| | Fitting | | | | | Guinier | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Time (min) | 4 | 8 | 16 | 32 | 64 | 4 | 8 | 16 | 32 | 64 |
| $R_g$ (nm) | – | 20 | (32)[a] | 45 | 68 | – | 19 | (32)[a] | 38 | 60 |
| $h$ (nm) | – | 3 | (4)[a] | 6 | 8 | – | 3 | (5)[a] | 5 | 7 |
| $L$ (nm) | – | 6 | (10)[a] | 14 | 21 | – | 6 | (10)[a] | 11 | 18 |

[a] Estimated values based on the same method. Because of the possible faulty data set (see text) these values should be treated with caution.

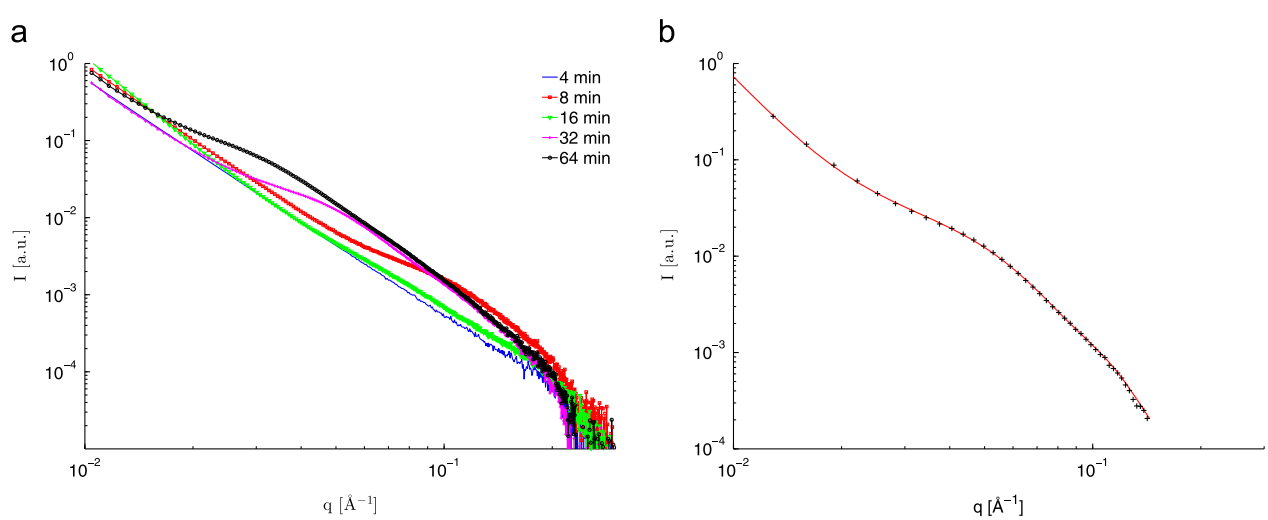


**Fig. 3.** Radially averaged intensity as a function of the scattering vector. (a) illustrates recorded data for five ageing times at 760 °C. Each set of ageing time data was fitted using SASfit to obtain parameters. (b) shows the fit (full line) for the 32 min ageing time data sample (marks). For clearness, the number of data points are reduced. Further details appear in the text.

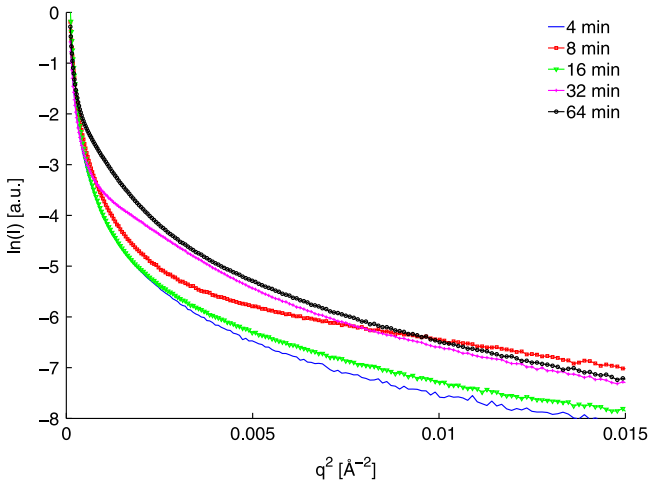


**Fig. 4.** Guinier representation of radially averaged SAXS data for five ageing times at 760 °C.

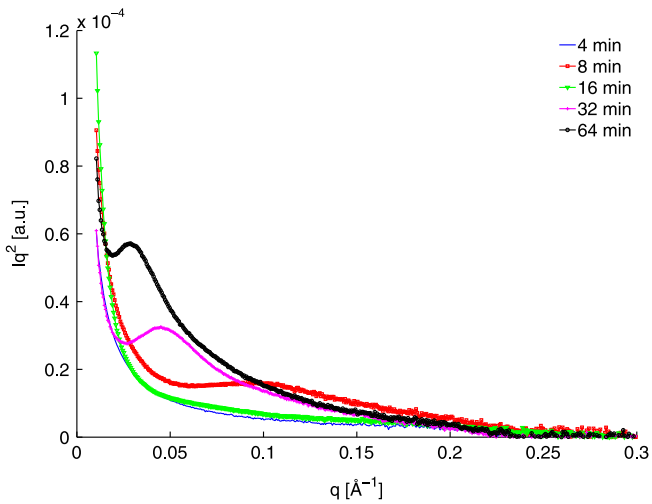


**Fig. 5.** Kratky representation of radially averaged SAXS data for five ageing times at 760 °C. The area below the curves represents the volume fraction of precipitates.

the placement and width strongly correlate with $R_g$ and the number of precipitates. The area under the curve represents the volume fraction. From the figure it is clear that the volume fraction increases with increasing ageing time.

The target area illuminated by the beam comprises a disperse distribution of precipitate sizes. Using Eq. (1), together with an applied log-normal distribution function, the precipitate size distribution can be determined. Fig. 6 illustrates the particle size distribution obtained for three ageing times (8, 32 and 64 min). For a short ageing time, the particle size distribution is almost mono-dispersive. With increasing time, the distributions become more dispersive and the mean particle size also increases.

### 4.3. Numeric comparison

The data from the current study are compared with previous measurements, which are shown in Fig. 7. The precipitate size in the early stages of nucleation and growth tend to follow previously established trends. The measured precipitate sizes for different ageing times are important information when developing a material model, accounting for nucleation, growth and coarsening of precipitates. A numerical model simulating the evolution of particle diameter and height is also included in Fig. 7. The model is based on the classical nucleation, growth and coarsening theory;

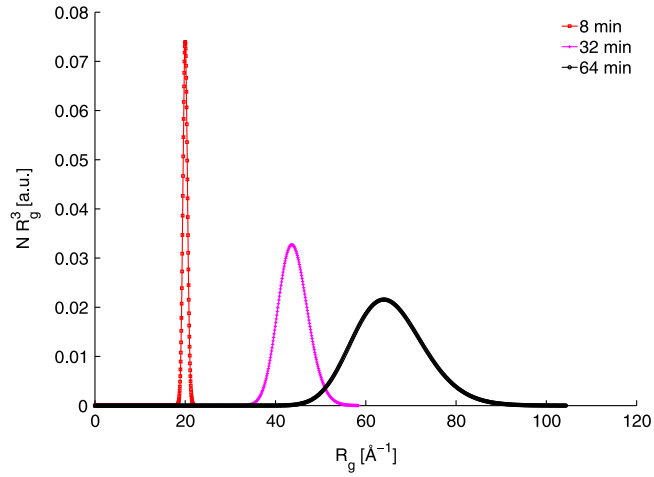


**Fig. 6.** Illustration of the particle size distribution. As can be seen from the figure, increasing ageing time results in: a broader width of the curve, indicating a larger particle size distribution, and a decreasing peak value, representing fewer precipitates due to Ostwald ripening.

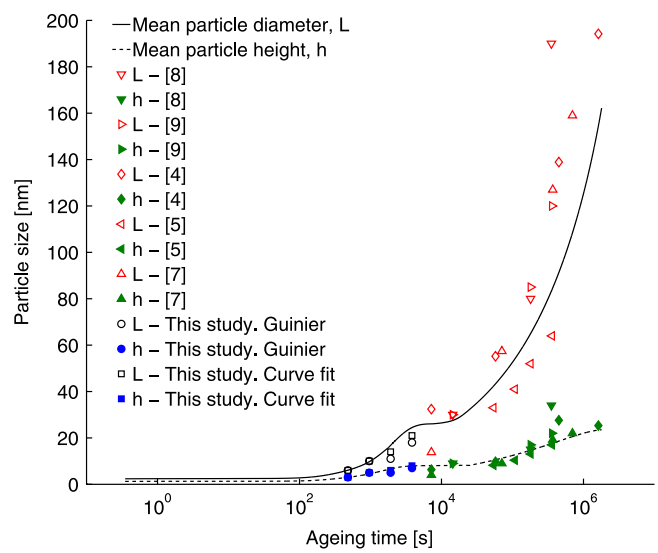


**Fig. 7.** Comparison of particle size evolution for this study and previous published results (see legend). The new measured data points tend to follow the previously established trend. The full line is a numerical comparison. Details of the model can be found in [37].

details of the model can be found elsewhere [12,37]. The precipitate material model can be coupled to a physically based, or dislocation density based, material model allowing the increase in strengthening during thermal treatment of a precipitate hardening alloy to be simulated [37,38].

## 5. Conclusions

The precipitate size in alloy IN718 has been investigated in the early stage of nucleation and growth using SAXS and Vickers microhardness measurements. Two different methods have been used to extract the radius of gyration from the SAXS data. No significant differences were observed between the two methods used.

Comparisons with earlier measurements reveal the evolution of precipitate size follows a similar trend, which is illustrated in Fig. 7. Data from the current study populate the early region in the figure, extending the known trend of precipitate evolution into a previously unmeasured region. To the authors' knowledge, these are the first measurements in the very early stags of nucleation

( < 10 min). It should be noted that the ageing temperature for this study was 10 °C higher than data from the comparisons. However, this is not expected to affect the general trend. The results from the SAXS measurements contain information on particle size and distribution. It can be seen from simulations that both the size and distribution increase for longer ageing times.

From the SAXS raw data, the variation of intensity as a function of the azimuthal angle are plotted, see Fig. 1(b). The variation is believed to arise from the processing of the material. This is studied by subtracting the azimuthal curve from the material aged for 4 min, which results in no clear evidence of anisotropy in the scatter. Therefore, it is concluded the precipitates are randomly oriented in the volume illuminated by the x-ray beam.

From the microhardness measurements shown in Fig. 2, an increase in hardness is revealed in the very early stage of ageing (~60% increase over the first 4 min at 760 °C). This suggests that an increase in hardness occurs when the precipitate is in its nucleation stage. This is further emphasised by the results from the SAXS measurements, where no scattering in the $q$-range could be seen before 8 min of ageing at 760 °C. This indicates the importance of making further studies for shorter ageing times, however for this purpose using the SAXS technique, a higher $q$-value on the beam set-up is needed.

The results from this work will be used as an important tool for developing new and accurate material models, which allow the increased yield strength for different ageing times and temperatures to be predicted. The model will be used, for example, to predict the precipitate sizes and their distribution inside a weld nugget and in the heat affected zone [39–41].

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